

AMGUN', Russian Federation

WMO: 314840

Lat: 51.417N

Lon: 135.083E

Elev: 269

StdP: 98.14

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-31.5	-36.7	0.1	-33.2	-34.6	0.1	-30.9	7.5	-20.8	6.2	-21.9	0.5	270	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.0	20.2	28.0	19.3	26.0	18.4	21.7	27.3	20.6	25.8	19.6	24.2	1.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.2	24.2	18.9	14.2	23.1	17.9	13.3	22.0	65.1	27.2	61.1	26.0	57.5	24.3	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.0	4.2	DB	-38.3	33.9	3.7	1.9	-41.0	35.3	-43.2	36.3	-45.3	37.4	-48.0	38.7	(4)
(5)			WB	-38.4	24.3	3.7	1.4	-41.1	25.4	-43.3	26.2	-45.3	27.0	-48.0	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-20.8	-17.0	-8.8	1.4	9.3	15.6	19.1	17.9	11.4	2.4	-9.9	-19.2	(6)	
	DBStd	14.92	5.14	4.78	5.23	4.43	4.53	4.11	3.44	3.24	3.80	4.22	5.80	5.27	(7)	
	HDD10.0	4398	954	757	583	261	69	3	0	1	28	239	597	906	(8)	
	HDD18.3	6729	1212	990	842	507	281	103	32	48	208	495	847	1164	(9)	
	CDD10.0	822	0	0	0	3	48	171	283	244	71	2	0	0	(10)	
	CDD18.3	113	0	0	0	0	1	21	56	33	1	0	0	0	(11)	
	CDH23.3	1148	0	0	0	2	51	287	512	274	22	1	0	0	(12)	
	CDH26.7	338	0	0	0	0	11	89	174	63	2	0	0	0	(13)	
(14) Wind	WSAvg	1.4	1.4	1.4	1.4	1.4	1.3	1.1	1.0	1.1	1.3	1.6	1.8	1.7	(14)	
(15) Precipitation	PrecAvg	621	18	11	18	39	80	83	119	107	66	41	21	21	(15)	
	PrecMax	798	58	41	62	98	140	172	242	265	130	117	51	85	(16)	
	PrecMin	358	0	1	2	9	28	24	4	32	20	7	1	1	(17)	
	PrecStd	117	16	9	15	24	33	34	51	57	32	29	15	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.0	-0.9	8.9	21.2	28.5	32.1	33.9	31.0	25.8	19.3	6.8	-3.3	(19)	
		MCWB	-6.3	-4.5	2.6	9.9	14.9	19.5	21.9	21.5	17.2	10.7	2.2	-4.6	(20)	
	2%	DB	-8.3	-4.0	4.8	16.9	24.5	29.3	30.5	28.5	23.2	14.7	3.1	-6.8	(21)	
		MCWB	-9.4	-6.6	-0.1	8.1	13.1	18.4	21.0	20.3	15.7	7.9	-0.3	-8.0	(22)	
	5%	DB	-10.6	-6.6	2.3	12.8	21.7	26.6	28.4	26.3	21.0	11.4	0.3	-9.2	(23)	
		MCWB	-11.6	-8.6	-1.7	5.6	11.9	17.3	20.3	19.4	14.4	5.8	-1.9	-10.3	(24)	
	10%	DB	-12.7	-8.9	0.0	9.6	18.9	24.1	26.1	24.4	18.5	8.9	-1.5	-11.3	(25)	
		MCWB	-13.6	-10.4	-3.1	3.9	10.8	16.7	19.6	18.4	13.1	4.5	-3.3	-12.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.3	-4.3	2.6	11.2	15.9	21.6	23.7	23.1	18.5	11.1	2.6	-4.3	(27)	
		MCDB	-5.1	-1.2	8.5	19.0	25.6	27.6	30.2	29.1	23.9	18.3	6.3	-3.2	(28)	
	2%	WB	-9.4	-6.5	0.1	8.4	14.1	19.7	22.2	21.4	16.7	8.6	0.0	-7.9	(29)	
		MCDB	-8.4	-4.1	4.3	16.5	22.2	26.5	27.8	26.5	21.5	13.3	2.1	-6.9	(30)	
	5%	WB	-11.6	-8.5	-1.4	6.0	12.9	18.4	21.1	20.3	15.4	6.6	-1.6	-10.2	(31)	
		MCDB	-10.7	-6.7	2.0	12.6	19.6	24.8	26.6	24.5	19.4	10.5	0.2	-9.2	(32)	
	10%	WB	-13.6	-10.4	-2.9	4.0	11.5	17.2	20.2	19.2	14.0	5.0	-3.4	-12.3	(33)	
		MCDB	-12.8	-8.9	-0.3	9.3	17.8	22.6	25.0	23.1	17.3	8.3	-1.5	-11.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	14.0	13.4	11.6	12.2	12.0	10.6	10.5	11.9	10.1	9.0	9.4	(35)	
		MCDBR	11.9	14.9	14.8	17.5	18.7	17.4	15.8	14.5	16.3	15.1	10.1	9.3	(36)	
		MCWBR	10.6	12.6	10.5	10.0	9.0	7.9	7.3	7.2	9.0	8.9	7.4	8.2	(37)	
	5% WB	MCDBR	11.5	14.6	14.0	16.8	16.4	15.1	13.2	12.4	14.3	14.3	9.5	9.0	(38)	
		MCWBR	10.3	12.3	10.0	9.9	8.5	7.5	6.8	6.7	8.7	8.8	7.2	8.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.261	0.295	0.367	0.475	0.550	0.496	0.490	0.416	0.372	0.374	0.307	0.260	(40)	
	taud		2.129	2.057	1.890	1.768	1.774	2.029	2.099	2.323	2.369	2.145	2.096	2.102	(41)	
	Ebn at Noon		742	809	802	753	718	769	766	813	810	703	655	666	(42)	
	Edh at Noon		74	107	155	199	208	163	150	114	97	99	77	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.53	2.72	4.32	5.14	4.86	5.26	4.84	4.30	3.37	2.14	1.47	1.18	(44)	
	RadStd		0.04	0.08	0.17	0.44	0.54	0.61	0.47	0.43	0.32	0.18	0.10	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page